

The examination is being carried out on the **following application documents**

Description, Pages

1-25 as published

Claims, Numbers

1-15 as published

1 Citations

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 WO 2012/032544 A2 (LAILA NUTRACEUTICALS [IN]; GOKARAJU GANGA RAJU [IN]; GOKARAJU RAMA RAJ) 15 March 2012 (2012-03-15)

D2 VAIDYA PANDITA HARIPRAPANNA JI: "Loharasayanam (Trayodasam)",
KEY ATTRIBUTES OF TKDL VS/1022, 12 August 2009 (2009-08-12), -
12 August 2009 (2009-08-12), pages 1-4, XP003025374,

D3 "Arq Barai Bars",
TKDL,, 1 January 1909 (1909-01-01), XP003029561,

D4 WO 2011/083397 A1 (HIMALAYA GLOBAL HOLDINGS LTD [AE];
PARAMESH RANGESH [IN]; BABU UDDAGIR) 14 July 2011
(2011-07-14)

D5 NAITHANI ET AL: "Decline in antioxidant capacity of Indian herbal teas
during storage and its relation to phenolic content",
FOOD RESEARCH INTERNATIONAL, ELSEVIER APPLIED SCIENCE,
BARKING, GB,
vol. 39, no. 2, 1 March 2006 (2006-03-01), pages 176-181,
XP005183541,
ISSN: 0963-9969, DOI: 10.1016/J.FOODRES.2005.07.004

- D6 WO 2005/094854 A1 (TORBANGUN PTY LTD [AU]; WAHLQVIST MARK LAWRENCE [AU]; WATTANAPENPAIBOO) 13 October 2005 (2005-10-13)
- D7 OBEN JULIUS E ET AL: "The effect of Cissus quadrangularis (CQR-300) and a Cissus formulation (CORE) on obesity and obesity-induced oxidative stress",
LIPIDS IN HEALTH AND DISEASE, BIOMED CENTRAL, LONDON, GB,
vol. 6, no. 1, 4 February 2007 (2007-02-04), page 4, XP021024438,
ISSN: 1476-511X, DOI: 10.1186/1476-511X-6-4
- D8 REDDY 1 K N ET AL: "Traditional knowledge on wild food plants in Andhra Pradesh",
INDIAN JOURNAL OF TRADITIONAL KNOWLEDGE, RESOURCES,
NEW DELHI, NEW DELHI - INDIA,
vol. 6, no. 1, 1 January 2007 (2007-01-01), pages 223-229,
XP018021499,
ISSN: 0972-5938
- D9 UPADHYAY B ET AL: "Ethnomedicinal and ethnopharmaco-statistical studies of Eastern Rajasthan, India",
JOURNAL OF ETHNOPHARMACOLOGY, ELSEVIER IRELAND LTD,
IE,
vol. 129, no. 1, 4 May 2010 (2010-05-04), pages 64-86, XP027013335,
ISSN: 0378-8741
[retrieved on 2010-03-07]
- D10 DMITRIY OBOLSKIY ET AL: "Garcinia mangostana L.: a phytochemical and pharmacological review",
PHYTOTHERAPY RESEARCH.,
vol. 23, no. 8, 1 August 2009 (2009-08-01), pages 1047-1065,
XP055314186,
GB
ISSN: 0951-418X, DOI: 10.1002/ptr.2730
- D11 US 2005/266018 A1 (BOREYKO BENSON K [US] ET AL) 1 December 2005 (2005-12-01)
- D12 WO 2011/068814 A1 (COLGATE PALMOLIVE CO [US]; TRIVEDI HARSH M [US]; GITTINS ELIZABETH K []) 9 June 2011 (2011-06-09)

D13 US 2009/252758 A1 (MAZED MOHAMMAD A [US] ET AL) 8 October 2009 (2009-10-08)

2 **Unity (Art. 82 EPC)**

2.1 The application lacks unity within the meaning of Article 82 EPC.

2.2 The problem to be solved by the present invention may be regarded as finding an alternative synergistic composition in regard to WO2012/032544 (D1) for enhancing physical performance and energy levels

2.3 The single general concept linking together the independent claims 1,2,14 and 15 is the following: plant extract or fraction as natural energy enhancer.

This single general concept is not novel / inventive (see eg WO2012/032544).

2.4 Hence, the following separate inventions or groups of inventions are not so linked as to form a single general inventive concept:

(1) synergistic composition comprising at least extracts or fractions derived from *Sphaeranthus indicus* and one of *Coleus aromaticus*, *Cissus quadrangularis*, *Curcuma longa*, *Citrullus lanatus*, *Ocimum sanctum* and *Cinnamomum tamala* as energy enhancer and for improving muscle strength, muscle mass and mental alertness

(2) synergistic composition comprising at least extracts or fractions derived from *Coleus aromaticus* and one of *Cissus quadrangularis*, *Curcuma longa*, *Garcinia mangostana*, *Citrullus lanatus*, *Ocimum sanctum* and *Cinnamomum tamala* as energy enhancer and for improving muscle strength, muscle mass and mental alertness

(3) synergistic composition comprising at least extracts or fractions derived from *Cissus quadrangularis* and one of *Curcuma longa*, *Garcinia mangostana*, *Citrullus lanatus*, *Ocimum sanctum* and *Cinnamomum tamala* as energy enhancer and for improving muscle strength, muscle mass and mental alertness

(4) synergistic composition comprising at least extracts or fractions derived from *Curcuma longa* and one of *Garcinia mangostana*, *Citrullus lanatus*, *Ocimum sanctum* and *Cinnamomum tamala* as energy enhancer and for improving muscle strength, muscle mass and mental alertness

(5) synergistic composition comprising at least extracts or fractions derived from *Garcinia mangostana* and one of *Citrullus lanatus*, *Ocimum sanctum* and *Cinnamomum tamala* as energy enhancer and for improving muscle strength, muscle mass and mental alertness

(6) synergistic composition comprising at least extracts or fractions derived from *Citrullus lanatus* and one of *Ocimum sanctum* and *Cinnamomum tamala* as energy enhancer and for improving muscle strength, muscle mass and mental alertness

(7) synergistic composition comprising at least extracts or fractions derived from *Ocimum sanctum* and *Cinnamomum tamala* as energy enhancer and for improving muscle strength, muscle mass and mental alertness

2.5 According to Article 82 and Rule 44 EPC a group of inventions is unitary if the group is linked so as to form a single general inventive concept. The link is present if there is a technical relationship between the inventions involving one or more of the same or corresponding special technical features. This special technical feature should define the contribution of the inventions over the prior art, i.e. be the reason for which the single general concept is novel and inventive. In the present case, the claims are directed to various plant combinations (e.g. *S. indicus* + *C. aromaticus* in composition 1, p. 15 versus *C. quadrangularis* + *C. longa* in composition 2,7,8, or 9, p. 15 and 16). The compositions as defined in inventions 1-7 below do not comprise one same plant / plant group. No further technical feature is apparent that could be regarded as a special technical feature establishing a technical relationship among the different inventions. Each of the inventions listed is a distinct invention, characterised by its own technical feature, namely the extract or fraction of each plant in the composition.

2.6 The application lacks unity within the meaning of Article 82 EPC.

2.7 The applicant is asked to state upon which invention searched further prosecution of this application should be based and to limit the application accordingly.

2.8 The subject-matter to be excised may be made the subject of one or more divisional applications according to Rule 36 EPC.

3 Treatment Method - Clarity - Support (Art. 84 EPC)

3.1 Claim 14

The subject-matter of claim 14 concerns a method of treatment of the human or animal body by surgery or therapy, which is excluded from patentability (Article 53(c) EPC). The claim as presently formulated is thus not allowable and should be either suppressed or reworded in the form of a non-therapeutic method since the application addresses the problem of providing energy in healthy mammals and not patients ("The non-medical / non-therapeutic use of ... for enhancing natural energy, ...").

This remark directed to claim 14 apply *mutatis mutandis* to claim 15.

3.2 Claim 15

Method claim 15 is not acceptable under Article 84 EPC. The therapeutic application is functionally defined by a mechanism of action which does not allow any practical application in the form of a defined, real treatment of a pathological condition (disease) (F-IV, 4.22 and G-II, 4.2).

Claim 15 also does not meet the requirements of Article 84 EPC in that the matter for which protection is sought is not defined:

(a) It is unclear how a biomarker can be "ameliorated". A biomarker either can be reduced or augmented.

(b) The terms "Muscle protein" and "Enhancing muscle cell proliferation" are unclear in correlation with "biomarker".

(c) Furthermore, the claim attempts to define the subject-matter in terms of the result to be achieved. In the absence of any knowledge of the claimed effect, either from the specification or from common general knowledge, the skilled person would not know before undertaking the laborious task of performing one or more screening assays how the given biomarkers would be affected by the various compositions.

Moreover, claim 15 is not supported by the description as required by Article 84 EPC, as its scope is broader than justified by the description and drawings. The scope of the claims must not be broader than is justified by the contribution to the state of the art.

3.3 Both claims 14 and 15 contain the term "as claimed in claims 1 to 3" which leads to unclarity and moreover is redundant since all features of claims 1 and 2 are already present in claims 14 and 15.

3.4 Claim 1 is unclear (as well as partly unsupported) in regard to the part "optionally containing at least one ingredient selected from plant, extracts, excipient, diluent or carrier" since this optional ingredient is not defined further than being a plant, extract, excipient, diluent or carrier. Any plant, extract, excipient, diluent or carrier can be included therein.

4 **Invention 1 (S. indicus combinations)**

4.1 **Novelty**

4.1.1 D1 claims (page 11, line 1 - line 14) a composition consisting of Sphaeranthus indicus derived herbal ingredient in combination with at least one ingredient selected from extract of Garcinia mangostana fruit rind, Ocimum basilicum, Zingiber officinalis, Tribulus terrestris, vitamins, amino acids, taurine, an extract of Ginkgo biloba, Rhodiola and an extract of guarana, optionally in combination with an excipient, diluent or carrier as a natural energy enhancer to provide an onset and steady maintenance of energy and mental alertness. Since the combination S. indicus and G. mangostana is excluded in independent claims 1 and 2, these claims can be regarded as novel. However, claim 14 (as well as 15) is novelty-destroyed by D1.

4.1.2 D2 discloses that Loharasäyanam (Trayodasam) is a therapeutic formulation consisting of useful parts of following ingredient(s) : Magnatite / Ferro-magnetic, Tinospora cordifolia Miers, Curcuma longa, Sphaeranthus indicus Linn., Tribulus terrestris Linn, and others. It is strength promoting. D2 destroys novelty of claims 1-3,5, and 8-15.

4.1.3 D3 discloses that Arq Barai Bars is a therapeutic single / compound formulation consisting of useful parts of following ingredient(s) : *Curcuma longa* (common turmeric, Turmeric), *Azadirachta indica* A. Juss. (neem), *Sphaeranthus indicus* Linn.). D3 destroys novelty of claims 1,2,5, 8-13, and 15.

4.1.4 D4 claims (claims 1-14; examples 21-24) a herbal composition for skin disorders comprising a blend of extract of *Psoralea corylifolia*, *Curcuma longa*, *Sphaeranthus indicus*, *Azadirachta indica* and *Andrographis paniculata*. Suitable dosage forms are tablet, capsule and syrup. D4 destroys novelty of claims 1,2,5, 8-13, and 15.

4.1.5 D5 discloses (abstract; table 1) herbal teas such as no. 3C comprising *Ocimum sanctum*, *Cinnamomum tamala*, *Viola odorata*, *Foeniculum vulgare*, *Cinnamomum zeylanicum*, *Zingiber officinale*, *Syzygium aromaticum*, *Myristica fragrance*, *Elettaria cardamomum*, *Piper nigrum*, *Cyperus rotundus*, *Tinospora cordifolia* and *Adhatoda vasica* or no. 6F comprising *Foeniculum vulgare*, *Cinnamomum zeylanicum*, *Trachyspermum ammi*, *Ocimum sanctum*, *Piper nigrum*, *Viola odorata*, *Mentha arvensis*, *Syzygium aromaticum*, *Zingiber officinale*, *Cinnamomum tamala*, *Bacopa monnieri*, *Glycyrrhiza glabra*, *Elettaria cardamomum*, *Withania somnifera*, *Rosa damascena*, *Santalum album* and *Terminalia arjuna*. These herbal teas are claimed to be antistress, energizer, stimulant, and memory enhancer. Since *S. indicus* is not mentioned, D5 is not novelty-destroying.

4.1.6 D6 claims (page 2, line 12 - line 26; claims 1-47) a nutritional composition comprising *Coleus amboinicus* which is synonymous of *Coleus aromaticus*. The Simalungun people believe *Coleus amboinicus* serves several purposes in a new mother, namely, stimulating and enhancing breast milk production, decreasing the risk of placental retention and restoring energy and strength lost during parturition. D6 as well is not novelty-destroying.

4.1.7 Hence, novelty cannot be acknowledged for claims 1-3,5, and 8-15.

4.2 Inventive step

4.2.1 The present application does not meet the requirements of Article 52(1) EPC because the subject-matter of independent claims 1,2,14, and 15 does not involve an inventive step within the meaning of Article 56 EPC.

- 4.2.2 D1 is considered to be the prior art closest to the subject-matter of these claims.
- 4.2.3 The subject-matter of the claims therefore differs from these known compositions for enhancing physical performance and energy levels in that the components *Coleus aromaticus*, *Cissus quadrangularis*, *Curcuma longa*, *Citrullus lanatus*, *Ocimum sanctum* and *Cinnamomum tamala* are not part of the combinations.
- 4.2.4 The problem to be solved by the present invention may therefore be regarded as finding alternative herbal compositions for the same use.
- 4.2.5 The solution proposed in claims 1,2,14, and 15 of the present application cannot be considered to involve an inventive step (Articles 52(1) and 56 EPC):
- 4.2.6 It is known from various documents such as D2, D5, and D6 that many plants can be used as energy enhancer, e.g. *Curcuma longa*, *Tinospora cordifolia*, *Sphaeranthus indicus*, *Tribulus terrestris* (D2); *Ocimum sanctum*, *Cinnamomum tamala*, *Foeniculum vulgare*, *Trachyspermum ammi*, *Piper nigrum*, *Mentha arvensis*, *Zingiber officinale*, etc (D5); *C. aromaticus* (D6). It would be obvious for the skilled in the art to combine these plant / plant extracts with *S. indicus* in order to arrive at the effect stated. The skilled person would expect at least an additional effect and would perform the standard "try and error" procedure with various combinations of above mentioned plants. No inventive effort is needed to arrive at the solution of providing these combinations as a solution to the problem of finding an energy enhancer.
- 4.2.7 However, the application has shown an effect for the combination *S. indicus* and *C. aromaticus* (combination 1, p. 16; tab. 7 and 8) when excluding the combinations containing *G. mangostana* (since they are disclaimed in claims 1 and 2). All other combinations with *S. indicus* have not been shown to lead to the claimed effect. In the case of combination 1, synergy has been shown. Hence, a claim referring to this combination only would be regarded as fulfilling the requirements of Article 56 EPC.

5 **Invention 3 (C. quadrangularis combinations)**

5.1 **Novelty**

- 5.1.1 D5 discloses (abstract; table 1) herbal teas such as no. 3C comprising Ocimum sanctum, Cinnamomum tamala, Viola odorata, Foeniculum vulgare, Cinnamomum zeylanicum, Zingiber officinale, Syzygium aromaticum, Myristica fragrance, Elettaria cardamomum, Piper nigrum, Cyperus rotundus, Tinospora cordifolia and Adhatoda vasica or no. 6F comprising Foeniculum vulgare, Cinnamomum zeylanicum, Trachyspermum ammi, Ocimum sanctum, Piper nigrum, Viola odorata, Mentha arvensis, Syzygium aromaticum, Zingiber officinale, Cinnamomum tamala, Bacopa monnieri, Glycyrrhiza glabra, Elettaria cardamomum, Withania somnifera, Rosa damascena, Santalum album and Terminalia arjuna. These herbal teas are claimed to be antistress, energizer, stimulant, and memory enhancer. Since Cissus quadrangularis is not mentioned, D5 is not novelty-destroying.
- 5.1.2 D7 discloses that Cissus quadrangularis promotes the increase of lean muscle mass. Furthermore, various formulations now are known which contain extracts of Cissus quadrangularis in combination with other compounds, used for the purpose of management of overweight and obesity, as well as complications resulting from these conditions.
- 5.1.3 D8 discloses in table 1 - listing wild plant species used as food by tribal people of Andhra Pradesh - that Cissus quadrangularis stem is crushed with Talati (Cleome gynandra) plant, turmeric and salt, and the mixture is taken orally. Hence, novelty for claims 1,2,4, 8-13, and 15 has to be denied.
- 5.1.4 D9 discloses (page 72, lines 5-6) that Cissus quadrangularis stem is made into paste, mixed with turmeric powder and castor oil and smeared on fractured bone for early cure. Like D8, D9 is novelty-destroying for claims 1,2,4, 8-13, and 15.
- 5.1.5 Hence, novelty cannot be acknowledged for claims 1,2,4, 8-13, and 15.

5.2 **Inventive step**

- 5.2.1 In case of novelty: The present application seems to meet partly the requirements of Article 52(1) EPC because the subject-matter of independent claims 1,2,14, and 15 seems to involve partly an inventive step within the meaning of Article 56 EPC.

- 5.2.2 D5 may be considered to be the prior art closest to the subject-matter of these claims.
- 5.2.3 The subject-matter of the claims therefore differs from these known compositions for enhancing physical performance and energy levels in that the components *Cissus quadrangularis*, *Curcuma longa*, *Citrullus lanatus*, *Ocimum sanctum* and *Cinnamomum tamala* are not part of the combinations.
- 5.2.4 The problem to be solved by the present invention may therefore be regarded as finding alternative herbal compositions for the same use.
- 5.2.5 It is known from various documents such as D2 and D5 that many plants can be used as energy enhancer, e.g. *Curcuma longa* (D2); *Ocimum sanctum*, *Cinnamomum tamala*, *Foeniculum vulgare*, *Trachyspermum ammi*, *Piper nigrum*, *Mentha arvensis*, *Zingiber officinale*, etc (D5). However, the use of *C. quadrangularis* as an energy enhancer has not been disclosed.
- 5.2.6 The application shows an effect for the combination *C. quadrangularis* and *C. longa* (compositions 2 as well as 7,8,9, p. 15-16; tab. 6, 7 and 8) whereby compositions 7,8,9 additionally contain *T. ammi*. All other combinations with *C. quadrangularis* have not been shown to lead to the claimed effect. In the case of combination 2, synergy has been shown. Hence, a claim referring to this combination only would be regarded as fulfilling the requirements of Article 56 EPC.
- 5.2.7 Regarding the use in improving muscle strength and muscle mass, D7 could be taken as closest prior art. D7 does not disclose a combination of *C. quadrangularis* with any of *Curcuma longa*, *Garcinia mangostana*, *Citrullus lanatus*, *Ocimum sanctum* and *Cinnamomum tamala*.
- 5.2.8 The problem to be solved by the present invention may therefore be regarded as finding alternative herbal compositions for the same use.
- 5.2.9 Since there is no prior art at hand combining the above mentioned plants for use in increasing muscle mass or teaching similar uses of these plants, and synergy has been shown for the combination *C. quadrangularis* and *C. longa* (as indicated in item 5.2.6), inventive step for this combination seems to be acknowledgeable.

6 Invention 5 (*G. mangostana* combinations)

6.1 Novelty

- 6.1.1 D1 claims (page 11, line 1 - line 14) a composition consisting of *Sphaeranthus indicus* derived herbal ingredient in combination with at least one ingredient selected from extract of *Garcinia mangostana* fruit rind, *Ocimum basilicum*, *Zingiber officinalis*, *Tribulus terrestris*, vitamins, amino acids, taurine, an extract of *Ginkgo biloba*, *Rhodiola* and an extract of guarana, optionally in combination with an excipient, diluent or carrier as a natural energy enhancer to provide an onset and steady maintenance of energy and mental alertness. Since the combinations *Garcinia mangostana* and one of *C. lanatus*, *O. sanctum*, and *C. tamala* are not disclosed, the claims can be regarded as novel.
- 6.1.2 D5 discloses (abstract; table 1) herbal teas such as no. 3C comprising *Ocimum sanctum*, *Cinnamomum tamala*, *Viola odorata*, *Foeniculum vulgare*, *Cinnamomum zeylanicum*, *Zingiber officinale*, *Syzygium aromaticum*, *Myristica fragrance*, *Elettaria cardamomum*, *Piper nigrum*, *Cyperus rotundus*, *Tinospora cordifolia* and *Adhatoda vasica* or no. 6F comprising *Foeniculum vulgare*, *Cinnamomum zeylanicum*, *Trachyspermum ammi*, *Ocimum sanctum*, *Piper nigrum*, *Viola odorata*, *Mentha arvensis*, *Syzygium aromaticum*, *Zingiber officinale*, *Cinnamomum tamala*, *Bacopa monnieri*, *Glycyrrhiza glabra*, *Elettaria cardamomum*, *Withania somnifera*, *Rosa damascena*, *Santalum album* and *Terminalia arjuna*. These herbal teas are claimed to be antistress, energizer, stimulant, and memory enhancer. Since *G. mangostana* is not mentioned, D5 is not novelty-destroying.
- 6.1.3 D10 discloses (page 1048, column 1, paragraph 3) that in Latin America a tea made from mangosteen fruits is used widely as a tonic for fatigue and low energy states.
- 6.1.4 D11 claims (paragraphs [0010] - [0012]; claims 1,21,24) a nutraceutical composition comprising mangosteen juice, mangosteen pericarp and edible cactus. Added to the nutraceutical composition can be one or more herbs, such as angelica, basil; a fruit and/or a vegetable such as watermelon (*C. lanatus*). Hence, claims 1-3, 8-13, and 15 are novelty-destroyed.
- 6.1.5 D12 claims (paragraph [0032] - paragraph [0036]; example 1) a composition containing a natural extract together with an extract from *Garcinia mangostana*. Particularly preferred extracts are *Zingiber officinale*, *Cinnamomum camphora*, *Cinnamomum verum*, *Curcuma longa*, *Ocimum*

sanctum, Foeniculum vulgare, Zingiber officinale. Since the combination G. mangostana with O. sanctum is disclosed, novelty for claims 1-3,8-13, and 15 has to be denied.

6.1.6 D13 discloses (paragraphs [0092] - [0094]; claim 1; example 1) the Mixture A comprising Mangosteen and Curcumin Powder. It is noted that Garcinia mangostana contains more than 40 different xanthones. They help the body run as a complete, interrelated and integrated system. They help the mind think better, adapt better, and give a sense of well-being (anti-depressant).

6.1.7 Hence, novelty cannot be acknowledged for claims 1-3, 8-13 and 15.

6.2 Inventive step

6.2.1 In case of novelty: The present application does not meet the requirements of Article 52(1) EPC because the subject-matter of independent claims 1,2,14, and 15 does not involve an inventive step within the meaning of Article 56 EPC.

6.2.2 D1 is considered to be the prior art closest to the subject-matter of these claims.

6.2.3 The subject-matter of the claims therefore differs from these known compositions for enhancing physical performance and energy levels in that the components Citrullus lanatus, Ocimum sanctum and Cinnamomum tamala are not part of the combinations.

6.2.4 The problem to be solved by the present invention may therefore be regarded as finding alternative herbal compositions for the same use.

6.2.5 The solution proposed in claims 1,2,14, and 15 of the present application cannot be considered to involve an inventive step (Articles 52(1) and 56 EPC):

6.2.6 G. mangostana is used as energy enhancer in traditional medicine (see D10) and in various compositions such as in D1 and D13 for enhancing physical performance as well as mental alertness and for generating a sense of well-being.

- 6.2.7 Moreover, it is known from various documents such as D5 that many plants can be used as energy enhancer, e.g. *Ocimum sanctum*, *Cinnamomum tamala*, *Foeniculum vulgare*, *Trachyspermum ammi*, *Piper nigrum*, *Mentha arvensis*, *Zingiber officinale*, etc (D5). It would be obvious for the skilled in the art to combine these plant / plant extracts with *G. mangostana* in order to arrive at the effect stated. The skilled person would expect at least an additional effect and would perform the standard "try and error" procedure with various combinations of above mentioned plants. No inventive effort is needed to arrive at the solution of providing these combinations as a solution to the problem of finding an energy enhancer.
- 6.2.8 The application shows an effect for the combination *G. mangostana* and *C. tamala* (composition 3, p. 15; tab. 7 and 8). All other combinations with *G. mangostana* have not been shown to lead to the claimed effect. In the case of combination 3, synergy has been shown. Hence, a claim referring to this combination only would be regarded as fulfilling the requirements of Article 56 EPC.